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Electronic package and manufacturing method thereof

Abstract

Provided is an electronic package providing a circuit structure having auxiliary circuit layers. Further, an electronic component is disposed on the circuit structure and electrically connected to the auxiliary circuit layers. In addition, an encapsulant covers the electronic component, and the circuit structure is disposed on the package substrate having a plurality of main circuit layers, such that the main circuit layers are electrically connected to the auxiliary circuit layers. As such, a number of layers of the auxiliary circuit layers is used to replace a layer number configuration of the main circuit layers.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of U.S. patent application Ser. No. 17/527,434, filed on Nov. 16, 2021, which is based upon and claims the right of priority to TW Patent Application No. 110138003, filed Oct. 13, 2021, both of which are incorporated herein by reference in their entirety for all purposes.

BACKGROUND

Technical Field

(1) The present disclosure relates to a semiconductor package, and more particularly, to an electronic package with a package module and a method for manufacturing the same.

Description of Related Art

(2) With the increase in the demand for functions and processing speed of electronic products, the semiconductor chip as the core component of the electronic product needs to have higher density electronic components and electronic circuits. Hence, the semiconductor chip will generate a larger amount of heat when it is in operation.

(3) Therefore, in order to quickly dissipate heat to the outside, the industry usually configures a heat sink or a heat spreader in the semiconductor package. The heat sink is usually bonded to the back of the chip by a heat-dissipating paste such that heat generated by the semiconductor chip can be dissipated by the heat-dissipating paste and the heat sink.

(4) As shown in FIG. 1, a conventional method for manufacturing a semiconductor package **1** is to first dispose a semiconductor chip **11** with an active surface **11a** thereof on a package substrate **10** in a flip-chip bonding manner (i.e., via conductive bumps **110** and an underfill **111**). A top plate **130** of a heat sink **13** is subsequently bonded onto a non-active surface **11b** of the semiconductor chip **11** by a heat-dissipating paste **12**, and supporting legs **131** of the heat sink **13** are disposed on the package substrate **10** via the adhesive layer **14**. During operation, heat generated by the semiconductor chip **11** is conducted to the top plate **130** of the heat sink **13** via the non-active surface **11b** and the heat-dissipating paste **12** to dissipate heat to the outside of the semiconductor package **1**.

(5) However, as the semiconductor chip **11** has more and more functional requirements, the number of input/output (I/O) pins thereof is also increasing. Therefore, the demand for the number of circuit layers of the package substrate **10** increases, but the yield of the package substrate **10** with multiple circuit layers (such as 16 layers or more) is low, resulting in a substantial increase in the overall cost of the semiconductor package **1**.

(6) Therefore, there is a need for a solution that addresses the above-mentioned issues of the prior art.

SUMMARY

(7) In view of the various above-mentioned shortcomings of the prior art, the present disclosure provides an electronic package, which includes: a package substrate having a plurality of main circuit layers; a package module disposed on the package substrate and including: a circuit structure disposed on the package substrate by a plurality of conductive components and having a plurality of auxiliary circuit layers electrically connected to the plurality of main circuit layers, wherein the circuit structure has a first surface and a second surface opposite to each other such that the second surface of the circuit structure is disposed on the package substrate via the plurality of conductive components, and a number of layers of the plurality of auxiliary circuit layers is used to replace the layer number configuration of the plurality of main circuit layers such that a number of layers of the plurality of main circuit layers is less than an original expected number of layers of the plurality of main circuit layers; an electronic component disposed on the first surface of the circuit structure and electrically connected to the plurality of auxiliary circuit layers, wherein the circuit structure extends out of sides of the electronic component; and an encapsulant formed on the first surface of the circuit structure to encapsulate the electronic component; and a packaging material formed between the package substrate and the package module to cover the plurality of

conductive components.

(8) The present disclosure further provides a method for manufacturing an electronic package, which includes: providing a circuit structure including a plurality of auxiliary circuit layers and a package substrate having a plurality of main circuit layers, wherein the circuit structure has a first surface and a second surface opposite to each other, and a number of layers of the plurality of auxiliary circuit layers is to replace a layer number configuration of the plurality of main circuit layers such that a number of layers of the plurality of main circuit layers is less than an original expected number of layers of the plurality of main circuit layers; disposing an electronic component on the first surface of the circuit structure and electrically connecting the electronic component with the plurality of auxiliary circuit layers, wherein the circuit structure extends out of sides of the electronic component; forming an encapsulant on the first surface of the circuit structure to encapsulate the electronic component to obtain a package module; disposing the second surface of the circuit structure of the package module on the package substrate having the plurality of main circuit layers via a plurality of conductive components, wherein the plurality of main circuit layers are electrically connected to the plurality of auxiliary circuit layers via the plurality of conductive components; and forming a packaging material on the package substrate, wherein the packaging material covers the plurality of conductive components.

(9) In the aforementioned electronic package and method for manufacturing the same, the circuit structure has a vertical projected area that is smaller than a vertical projected area of the package substrate.

(10) In the aforementioned electronic package and method for manufacturing the same, the plurality of auxiliary circuit layers have a single layer that replaces 2 to 4 layers of the plurality of main circuit layers.

(11) In the aforementioned electronic package and method for manufacturing the same, the package substrate includes a core layer and build-up structures disposed on opposite sides of the core layer, and each of the build-up structures includes a plurality of dielectric layers and the plurality of main circuit layers bonded to the plurality of dielectric layers. For example, the number of layers of the plurality of main circuit layers of the package substrate is at least ten.

(12) In the aforementioned electronic package and method for manufacturing the same, the package substrate has an area that is at least 60×60 mm.^{sup.2}

(13) In the aforementioned electronic package and method for manufacturing the same, the electronic component has a plurality of electrode pads electrically connected to the plurality of auxiliary circuit layers, and a distance between the electrode pads is less than or equal to 60 μm.

(14) In the aforementioned electronic package and method for manufacturing the same, the plurality of auxiliary circuit layers of the circuit structure facing a side of the package substrate have a plurality of electrical contact pads, and a distance between the electrical contact pads is 80 μm to 150 μm or 200 μm to 300 μm.

(15) The aforementioned electronic package and method for manufacturing the same further include disposing a plurality of ball pads on another side of the package substrate opposite to the package module, wherein a distance between the ball pads is 500 μm to 1,000 μm.

(16) The aforementioned electronic package and method for manufacturing the same further include disposing a heat sink on the package substrate for heat dissipation of the package module.

(17) As can be understood from the above, in the electronic package and method for manufacturing the same of the present disclosure, the number of layers of circuit layers of the high-density layout specifications required by the electronic package that is mainly configured in the circuit structure to reduce the demand for the number of layers of the circuit layers of the package substrate and to enable the number of layers of the auxiliary circuit layers and the number of layers of the main circuit layers have a complementary relationship. Therefore, compared with the prior art, the electronic package of the present disclosure uses the number of layers of the auxiliary circuit layers to replace a layer number configuration of the main circuit layers, such that the number of layers of

the main circuit layers is less than the original expected number of layers of the main circuit layers. As a result, since the number of layers of main circuit layers of the package substrate needed is greatly reduced, the yield of the package substrate can be improved, and the overall cost of the electronic package can be reduced.

(18) Further, the present disclosure uses a single auxiliary circuit layer to replace the multi-layer main circuit layer design such that the overall thickness of the electronic package can be relatively thin, and the volume of the electronic package of the present disclosure can be greatly reduced.

(19) Additionally, in the present disclosure, the package substrate has a core layer configuration, which can improve the overall structural strength such that the present disclosure can avoid the problem of warpage during the manufacturing process.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a schematic cross-sectional view of a conventional semiconductor package.

(2) FIGS. 2A to 2C and 2D-1 are schematic cross-sectional views illustrating a method for manufacturing an electronic package in accordance with the present disclosure.

(3) FIG. 2D-2 is a schematic cross-sectional view illustrating another method for manufacturing an electronic package of FIG. 2D-1.

(4) FIG. 2E is a schematic partial bottom view of FIG. 2D-1.

(5) FIG. 3 is a schematic cross-sectional view illustrating an electronic package in accordance with another embodiment of the present disclosure.

(6) FIG. 4 is a graph illustrating a cost-benefit comparison of the electronic package in accordance with the present disclosure.

DETAILED DESCRIPTION

(7) Implementations of the present disclosure are illustrated using the following embodiments. One of ordinary skill in the art can readily appreciate other advantages and technical effects of the present disclosure upon reading the content of this specification.

(8) It should be noted that the structures, ratios, sizes shown in the drawings appended to this specification are to be construed in conjunction with the disclosure of this specification in order to facilitate understanding of those skilled in the art. They are not meant, in any ways, to limit the implementations of the present disclosure, and therefore have no substantial technical meaning. Any modifications, changes or adjustments to the structures, ratio relationships or sizes, are to be construed as falling within the range covered by the technical content disclosed herein to the extent of not causing changes in the technical effects created and the objectives achieved by the present disclosure. Meanwhile, terms such as “above,” “first,” “second,” “a,” “an,” and the like recited herein are for illustrative purposes, and are not meant to limit the scope in which the present disclosure can be implemented. Any variations or modifications to their relative relationships, without changes in the substantial technical content, should also to be regarded as within the scope in which the present disclosure can be implemented.

(9) FIGS. 2A to 2D-1 are schematic cross-sectional views illustrating a method for manufacturing an electronic package 2 in accordance with the present disclosure.

(10) As shown in FIG. 2A, a circuit structure 20 is formed on a carrier 9. The circuit structure 20 is defined with a first surface 20a and a second surface 20b opposite to each other, and the circuit structure 20 is bonded onto the carrier 9 with the second surface 20b thereof.

(11) In an embodiment, the carrier 9 is a metal plate, a semiconductor wafer, or a glass plate, such as a temporary full-panel silicon (Si) wafer, on which a sacrificial release layer 90 is disposed such that the circuit structure 20 is bonded onto the sacrificial release layer 90.

(12) Further, the circuit structure 20 is, for example, a coreless substrate, which includes at least

one insulating layer **200** and at least one auxiliary circuit layer **201** combined with the insulating layer **200**, such as at least one fan-out redistribution layer (RDL). For example, the auxiliary circuit layer **201** at the first surface **20a** of the circuit structure **20** has a plurality of input/output (I/O) pins, and the pitch specification thereof is micron-level (μ -Pad).

(13) In addition, the material forming the auxiliary circuit layer **201** is copper, and the material forming the insulating layer **200** is a dielectric material such as polybenzoxazole (PBO), polyimide (PI) and prepreg (PP) or the like, or a solder-proof material such as solder mask or graphite.

(14) As shown in FIG. 2B, an electronic component **21** is connected onto the first surface **20a** of the circuit structure **20** via a plurality of conductive structures **22** such that the electronic component **21** is electrically connected to the auxiliary circuit layer **201**. Subsequently, an encapsulant **23** is formed on the first surface **20a** of the circuit structure **20** such that the encapsulant **23** encapsulates the electronic component **21** (e.g., in an encapsulating manner).

(15) In an embodiment, the electronic component **21** is an active component, a passive component, or a combination thereof. For example, the active component is a semiconductor chip, and the passive component is, for example, a resistor, a capacitor, or an inductor. For example, the electronic component **21** is a semiconductor chip, which has an active surface **21a** and a non-active surface **21b** opposite to each other. The active surface **21a** has a plurality of electrode pads **210**, which are disposed on the auxiliary circuit layer **201** of the first surface **20a** of the circuit structure **20** and electrically connected to the auxiliary circuit layer **201** via the plurality of conductive structures **22** such as solder materials, metal pillars or others in a flip-chip manner, and the conductive structures **22** are covered with insulating materials **24** such as an underfill or a non-conductive underfill film (NCF), etc. Alternatively, the electronic component **21** may be electrically connected to the auxiliary circuit layer **201** of the first surface **20a** of the circuit structure **20** by means of a plurality of bonding wires in a wire-bonding manner (not shown). Alternatively, the electronic component **21** may directly contact the auxiliary circuit layer **201** of the first surface **20a** of the circuit structure **20**. Therefore, there are many ways to electrically connect the electronic component **21** to the circuit structure **20**, but not limited to the above.

(16) Moreover, the distance d between the electrode pads **210** is less than or equal to $60\text{ }\mu\text{m}$, and the conductive structure **22** includes a conductor **220** bonded to the electrode pad **210** and a solder material **221** bonded to the conductor **220** and the auxiliary circuit layer **201**. For example, the conductor **220** may be a metal pillar such as a copper pillar.

(17) Besides, the encapsulant **23** is an insulating material, such as a polyimide (PI), a dry film, an encapsulation colloid such as epoxy resin, or molding compound. For example, the encapsulant **23** may be formed on the circuit structure **20** by means of liquid compound, injection, lamination or compression molding, etc.

(18) In addition, the insulating material **24** may be formed between the active surface **21a** of the electronic component **21** and the first surface **20a** of the circuit structure **20** to cover the conductive structures **22**, and then the encapsulant **23** covers the insulating material **24**. Alternatively, the insulating material **24** is free from being formed such that the encapsulant **23** directly covers the conductive structures **22**.

(19) As shown in FIG. 2C, the carrier **9** and the sacrificial release layer **90** thereon are removed to expose the second surface **20b** of the circuit structure **20**, and then a plurality of conductive components **25** (such as solder bumps) are formed on the second surface **20b** of the circuit structure **20**.

(20) In an embodiment, the auxiliary circuit layer **201** has a plurality of electrical contact pads **202** at the second surface **20b**, and the surface **202a** of the electrical contact pad **202** is flushed with the second surface **20b** of the circuit structure **20**. For example, the distance $t1$ between the electrical contact pads **202** is $80\text{ to }150\text{ }\mu\text{m}$ or $200\text{ to }300\text{ }\mu\text{m}$.

(21) Furthermore, an under bump metallurgy (UBM) (not shown) may be formed on the electrical contact pad **202** to facilitate the bonding of the conductive component **25**.

(22) As shown in FIG. 2D-1, a singulation process is performed along a cutting path S shown in FIG. 2C to obtain a plurality of package modules **2a**, and the circuit structure **20** extends out of sides **21c** of the electronic component **21**. Subsequently, the package module **2a** is disposed onto a package substrate **26** via the conductive components **25**, and a packaging material **27** (such as an underfill) is formed on the package substrate **26** such that the packaging material **27** covers the conductive components **25**. After that, a heat sink **28** is disposed on the package substrate **26** to cover the package module **2a**.

(23) In an embodiment, the area of the surface of the package substrate **26** connected to the package module **2a** is a large-size version, which is at least 60×60 mm^{sup.2} (a square with the same length L1 and width L2 as shown in FIG. 2E or a rectangle) such that the vertical projected area A1 of the circuit structure **20** is smaller than the vertical projected area A2 of the package substrate **26**, as shown in FIGS. 2D-1 and 2E. In addition, the package substrate **26** includes a core layer **26a** and build-up structures **26b**, **26c** disposed on opposite sides of the core layer **26a**, wherein the core layer **26a** has conductive vias **260**, and each of the build-up structures **26b**, **26c** has a dielectric body **262** composed of a plurality of dielectric layers and the main circuit layers **261** of the conductive vias **260** bonded to and electrically connected to the plurality of dielectric layers. For example, the material forming the main circuit layers **261** is copper, and the material forming the dielectric body **262** is a dielectric material such as polybenzoxazole (PBO), polyimide (PI) and prepreg (PP).

(24) Moreover, the heat sink **28** has a heat-dissipating body **280** and a plurality of supporting legs **281** extending downward from the edge of the heat-dissipating body **280**. The heat-dissipating body **280** is a type of heat sink, and the lower side of the heat-dissipating body **280** is bonded onto the encapsulant **23** of the package module **2a** by pressing a heat dissipation material **28a** such that the heat dissipation material **28a** is located between the heat-dissipating body **280** and the electronic component **21**, and the supporting legs **281** are bonded onto the package substrate **26** by an adhesive layer **28b**. Alternatively, the package module **2a** may be flattened by a flattening process (such as a grinding method) to make the non-active surface **21b** of the electronic component **21** flush with the surface of the encapsulant **23**, as shown in FIG. 2D-2. Consequently, the non-active surface **21b** of the electronic component **21** is exposed from the encapsulant **23** to contact the heat dissipation material **28a**. For example, the heat dissipation material **28a** is a thermal interface material (TIM), a solder material, a metal material or other thermally conductive materials.

(25) Furthermore, in another embodiment, as shown in FIG. 3 for the electronic package **3**, the heat sink **38** may also be a ring body, and the lower side of the heat sink **38** is bonded onto the package substrate **26** by the adhesive layer **28b**. In addition, the heat sink **38** has an opening **380** corresponding to the position of the package module **2a** such that the package module **2a** is exposed from the opening **380**, and can even protrude from the opening **380** as required. It should be understood that the non-active surface **21b** of the electronic component **21** does not need to be exposed from the encapsulant **23**.

(26) Besides, in the subsequent manufacturing process, the plurality of ball pads **263** of the main circuit layers **261** on the lower side of the package substrate **26** can be bonded with the solder balls **29** such that the electronic packages **2**, **3** are connected to an electronic device (such as a circuit board) (not shown) by the solder balls **29**. For example, the distance t2 between the ball pads **263** is 500 to 1000 μm .

(27) As such, by disposing the electronic component **21** on a circuit structure **20** in the form of a redistribution layer (RDL) to produce the package module **2a**, the electrode pad **210** of the electronic component **21** can be connected onto the package substrate **26** via the circuit structure **20**. Hence, the pitch specification of the input/output (I/O) pins (i.e., the positions corresponding to the conductive structures **22**) of the auxiliary circuit layers **201** on the first surface **20a** of the circuit structure **20** can match the input/output (I/O) pin pitch specification of the electronic

component **21** (for example, the distance $d \leq 60 \mu\text{m}$ between the electrode pads **210**). Additionally, the pitch specification of the input/output (I/O) pins (for example, the distance $t1$ between the electrical contact pads **202** is 80 to 150 μm or 200 to 300 μm) of the auxiliary circuit layers **201** on the second surface **20b** of the circuit structure **20** can match the pitch specification of the input/output (I/O) pins (i.e., the positions corresponding to the conductive components **25**) of the package substrate **26**. As a result, the package substrate **26** can be manufactured using existing processes (for example, the input/output [I/O] pin pitch corresponding to each conductive component **25** is made to be 130 μm), and can still be packaged with the electronic components **21** of advanced specifications to form the required multifunctional electronic packages **2, 3**, thereby facilitating the development of electronic products using the electronic packages **2, 3**.

(28) Furthermore, as the functional requirements of the electronic component **21** increase, the number of input/output (I/O) pins thereof also increases. In order to meet the number of input/output (I/O) pins, the conventional package substrate **10** needs to be designed with more than 16 layers before layout. Therefore, the number of layers of circuit layers of high-density layout specifications required by the electronic package **2, 3** are disposed in the circuit structure **20** of the package module **2a** (i.e., the number of layers of the auxiliary circuit layers **201**) so as to reduce the requirement for the number of layers of circuit layers (i.e., the number of layers of the main circuit layers **261**) of the package substrate **26**. Consequently, the number of layers of the auxiliary circuit layers **201** and the number of layers of the main circuit layers **261** are in a complementary relationship. Therefore, compared with the prior art, the number of layers of circuit layers of the package substrate **26** required for the electronic packages **2** and **3** of the present disclosure is relatively low (for example, the number of layers of the main circuit layers **261** is less than ten). In addition, except that the auxiliary circuit layers **201** are used to assist and meet the process capability of the package substrate **26** (for example, the input/output (I/O) pin pitch being 80 to 150 μm), the number of layers of the auxiliary circuit layers **201** can further replace more layers of the main circuit layers **261** such that the number of layers of the package substrate **26** is reduced. Moreover, the input/output (I/O) pin pitch thereof can be designed to a larger specification of 200 to 300 μm , which further improves the yield of the package substrate **26** and reduces the cost of the package substrate **26**, thereby reducing the overall cost of the electronic package **2, 3**.

(29) Additionally, since the RDL specification of the circuit structure **20** can produce a higher density pitch than the package substrate **26**, each auxiliary circuit layer **201** in the form of RDL can replace 2 to 4 layers of the main circuit layers **261**, and the configurations of the plurality of main circuit layers **261** and the auxiliary circuit layers **201** can meet the circuit functions required by the electronic component **21**. Thus, compared with the prior art, the electronic package **2, 3** of the present disclosure can be relatively thinner in overall thickness (or height) such that the volume of the electronic package **2, 3** can be reduced.

(30) For example, in view of the **16** circuit layers of the existing package substrate (i.e., each of the upper and lower surfaces of the core layer **26a** has one main circuit layer **261**, that is, there are two layers in total, and there are 7 main circuit layers **261** on the upper and lower sides of the core layer **26a** to form build-up structures **26b, 26c** such that the number of layers can be expressed as 7/2/7), if one auxiliary circuit layer **201** in the form of RDL replaces two main circuit layers **261** of the package substrate **26**, the number of layers of the main circuit layers **261** of the package substrate **26** is configured to become 6/2/6 when the circuit structure **20** has only one auxiliary circuit layer **201**. Alternatively, when the circuit structure **20** has three auxiliary circuit layers **201**, as shown in FIG. 2D-1, the number of layers of the main circuit layers **261** of the package substrate **26** is configured to become 4/2/4, and so on.

(31) Similarly, if one auxiliary circuit layer **201** in the form of RDL replaces four main circuit layers **261** of the package substrate **26**, the number of layers of the main circuit layers **261** of the package substrate **26** is configured to become 5/2/5 when the circuit structure **20** has only one auxiliary circuit layer **201**. Alternatively, when the circuit structure **20** has two auxiliary circuit

layers **201**, the number of layers of the main circuit layers **261** of the package substrate **26** is configured to become 3/2/3, and so on.

(32) It should be understood that if the number of layers of the auxiliary circuit layers **201** is greater, the number of layers of the main circuit layers **261** of the package substrate **26** is less.

(33) Preferably, since a single auxiliary circuit layer **201** can replace 2 to 4 layers of the main circuit layers **261**, compared with the package substrate **26** that originally requires more than 16 layers, the circuit structure **20** that replaces some layers of the main circuit layers **261** is more cost-effective. That is, the more the number of layers of circuit layers originally required for the package substrate **26** is (as shown in FIG. 4, there are more than 16 layers or even more 18 layers), the better the cost-effectiveness of the overall package structure is after the circuit structure **20** is improved (as shown in FIG. 4, the cost is reduced by 11%, or even reduced by 18%). Conversely, if the number of layers of circuit layers originally required for the package substrate **26** is less than 16 layers, as shown in FIG. 4, less than 12 layers, or even less than 10 layers, the cost can only be reduced by 3%, or even increased (10 layers as shown in FIG. 4). It should be understood that the circuit structure **20** that replaces more layers of the main circuit layers **261** is not better, instead, it needs to cooperate with the process capability, structural strength or other specification constraints and cost calculations to obtain the optimal layer allocation.

(34) In addition, the manufacturing method of the present disclosure is mainly applied to the demand of high number of input/output (I/O) pins. Hence, the volume of the electronic package needs to be designed to be larger (for example, the area thereof is at least 60×60 mm.sup.2). If a coreless substrate is used in the manufacturing process, warpage is likely to occur. As such, with the configuration of the package substrate **26** having the core layer **26a**, the overall structural strength of the electronic packages **2, 3** can be improved to avoid the problem of warpage of the electronic packages **2, 3** due to large dimensions during manufacturing processes.

(35) The present disclosure further provides an electronic package **2, 3**, including: a package substrate **26** having a plurality of main circuit layers **261**, a package module **2a** disposed on the package substrate **26** and a packaging material **27**. The package module **2a** includes a circuit structure **20** disposed on the package substrate **26**, at least one electronic component **21** disposed on the circuit structure **20** and an encapsulant **23** disposed on the circuit structure **20**.

(36) The circuit structure **20** has a plurality of auxiliary circuit layers **201** electrically connected to the plurality of main circuit layers **261**. Besides, the circuit structure **20** has a first surface **20a** and a second surface **20b** opposite to each other such that the second surface **20b** of the circuit structure **20** has a plurality of conductive components **25** which are disposed on the package substrate **26**, and the number of layers of the auxiliary circuit layers **201** is used to replace the layer number configuration of the main circuit layers **261** such that the number of layers of the main circuit layers **261** is less than the original expected number of layers of the main circuit layers **261**.

(37) The electronic component **21** is disposed on the first surface **20a** of the circuit structure **20** and electrically connected to the auxiliary circuit layer **201**, wherein the circuit structure **20** extends out of the sides **21c** of the electronic component **21**.

(38) The encapsulant **23** is disposed on the first surface **20a** of the circuit structure **20** to encapsulate the electronic component **21**.

(39) The packaging material **27** is formed between the package substrate **26** and the package module **2a** to cover the plurality of conductive components **25**.

(40) In an embodiment, the vertical projected area **A1** of the circuit structure **20** is smaller than the vertical projected area **A2** of the package substrate **26**.

(41) In an embodiment, the single layer of the plurality of auxiliary circuit layers **201** can replace 2 to 4 layers of the main circuit layers **261**.

(42) In an embodiment, the package substrate **26** includes a core layer **26a** and build-up structures **26b, 26c** disposed on opposite sides of the core layer **26a**, and each of the build-up structure **26b, 26c** includes a dielectric body **262** composed of a plurality of dielectric layers and the plurality of

main circuit layers **261** bonded to the plurality of dielectric layers. For example, the number of layers of the main circuit layers **261** of the package substrate **26** is at least ten.

(43) In an embodiment, the area of the surface of the package substrate **26** connected to the package module **2a** is at least 60×60 mm.^{sup.2} (a square with the same length **L1** and width **L2** as shown in FIG. 2E or a rectangle).

(44) In an embodiment, the electronic component **21** has a plurality of electrode pads **210** electrically connected to the auxiliary circuit layer **201**, and the distance **d** between the electrode pads **210** is less than or equal to $60 \mu\text{m}$.

(45) In an embodiment, the auxiliary circuit layer **201** of the circuit structure **20** facing the side of the package substrate **26** has a plurality of electrical contact pads **202**, and the distance **t1** between the electrical contact pads **202** is 80 to $150 \mu\text{m}$ or 200 to $300 \mu\text{m}$.

(46) In an embodiment, a plurality of ball pads **263** are disposed on another side of the package substrate **26** opposite to the package module **2a**, and the distance **t2** between the ball pads **263** is 500 to $1000 \mu\text{m}$.

(47) In an embodiment, the electronic package **2, 3** further includes a heat sink **28, 38** disposed on the package substrate **26** for heat dissipation of the package module **2a**.

(48) In summary, the electronic package and the manufacturing method thereof according to the present disclosure utilize the configuration of the circuit structure to enable the electronic component to be disposed on the package substrate according to the specification requirements. Therefore, the package substrate can be manufactured by using the existing manufacturing process to encapsulate electronic components of various specifications into the required multifunctional electronic package so as to facilitate the development of electronic products using the electronic package.

(49) Moreover, the number of layers of circuit layers of high-density layout specifications required by the electronic package is configured in the circuit structure to reduce the demand for the number of layers of circuit layers of the package substrate. Therefore, since the number of layers of circuit layers of the package substrate required for the electronic package of the present disclosure is greatly reduced, the yield of the package substrate can be improved, and the overall cost of the electronic package can be reduced.

(50) Further, a single auxiliary circuit layer can replace multiple main circuit layers such that the overall thickness of the electronic package of the present disclosure can be relatively thin, and the volume of the electronic package can be reduced.

(51) Additionally, the package substrate has a core layer configuration, which improves the overall structural strength of the electronic package such that the present disclosure can avoid the problem of warpage during the manufacturing process.

(52) The above embodiments are set forth to illustrate the principles of the present disclosure, and should not be interpreted as to limit the present disclosure in any way. The above embodiments can be modified by one of ordinary skill in the art without departing from the scope of the present disclosure as defined in the appended claims.

Claims

1. An electronic package, comprising: a package substrate having a plurality of main circuit layers; a package module disposed on the package substrate and including: a circuit structure disposed on the package substrate and having a plurality of auxiliary circuit layers electrically connected to the plurality of main circuit layers, wherein the circuit structure has a first surface and a second surface opposite to each other such that the second surface of the circuit structure is disposed on the package substrate, and a number of layers of the plurality of auxiliary circuit layers is used to replace a layer number configuration of the plurality of main circuit layers such that a number of layers of the plurality of main circuit layers is less than an original expected number of layers of the

plurality of main circuit layers, and a sum of the number of layers of the plurality of auxiliary circuit layers and the number of layers of the plurality of main circuit layers is at least ten; an electronic component disposed on the first surface of the circuit structure and electrically connected to the plurality of auxiliary circuit layers, wherein the circuit structure has a vertical projected area smaller than a vertical projection area of the package substrate; and an encapsulant formed on the first surface of the circuit structure to encapsulate the electronic component, wherein the plurality of auxiliary circuit layers have a single layer replacing 2 to 4 layers of the plurality of main circuit layers, and the original expected number of layers of the plurality of main circuit layers is at least sixteen.

2. The electronic package of claim 1, wherein the circuit structure is disposed on the package substrate by a plurality of conductive components.

3. The electronic package of claim 1, wherein the second surface of the circuit structure has a plurality of conductive components.

4. The electronic package of claim 1, wherein the circuit structure extends out of sides of the electronic component.

5. The electronic package of claim 1, further comprising a packaging material formed between the package substrate and the package module to cover a plurality of conductive components.

6. The electronic package of claim 1, wherein the package substrate includes a core layer and build-up structures disposed on opposite sides of the core layer, and each of the build-up structures includes a plurality of dielectric layers and the plurality of main circuit layers bonded to the plurality of dielectric layers.

7. The electronic package of claim 6, wherein the number of layers of the plurality of main circuit layers of the package substrate is at least ten.

8. The electronic package of claim 1, wherein the package substrate has an area of at least 60×60 mm.^{sup.2}.

9. The electronic package of claim 1, wherein the electronic component has a plurality of electrode pads electrically connected to the plurality of auxiliary circuit layers, and a distance between the electrode pads is less than or equal to 60 μm.

10. The electronic package of claim 1, wherein the plurality of auxiliary circuit layers of the circuit structure facing a side of the package substrate have a plurality of electrical contact pads, and a distance between the electrical contact pads is 200 μm to 300 μm or 80 μm to 150 μm.

11. The electronic package of claim 1, further comprising a plurality of ball pads disposed on another side of the package substrate opposite to the package module, wherein a distance between the ball pads is 500 μm to 1,000 μm.

12. The electronic package of claim 1, further comprising a heat sink disposed on the package substrate for heat dissipation of the package module.
